Closing the Loop: Assessing the Efficacy of Medication Discontinuation Practices in Electronic Health Records

Ann Mathew, Allison B. McCoy, PhD
Computer Science & Climate and Environmental Studies, Vanderbilt University
Department of Biomedical Informatics, Vanderbilt University Medical Center

Background

Electronic health records (EHRs) are widely regarded to improve accuracy in documenting patient data, but there is limited information on the accuracy of the current medication list. While initiation of outpatient prescriptions is well-documented, there may be inconsistencies or inaccuracies when a patient is no longer taking a medication, in part due to varying clinical workflows involved in any given patient encounter. Medications can be requested to be removed by patients using the patient portal, marked as not taking or removed by nurses and medical assistants during visit intake, or marked as not taking or discontinued by physicians and advanced practice providers during a visit. Inaccuracies in medication lists could result in subsequent errors in clinical decision support alerts or research reliant on secondary medication data. We sought to identify the prevalence and impact of this issue in the medication lists for patients at Vanderbilt University Medical Center (VUMC).

Methods

We extracted medication data from the EHR for all patient encounters between June 2nd, 2024 and June 6th, 2024. We carried out a descriptive analysis to identify patterns in medication removal documentation. We also collected qualitative data from three physicians who interact directly with the medication list during patient encounters to improve understanding of the medication discontinuation workflow.

Results

Among 37,628 analyzed patient encounters, 35.25% (13,245 encounters) involved instances of medication discontinuation by physicians. Across all encounters, 7,315 medications were marked as not being taken either through the patient portal or during intake. Of these, only 2,593 medications (35.45%) remained on the medication list. In addition, 10,746 (28.56%) of 37,628 patient encounters involved the removal of medications from the EHR without any requests from MyChart or intake providers. While the majority of medications ordered by a provider from a given specialty were discontinued by a visit provider of the same specialty, some visit providers discontinued medications ordered from other specialties at almost the same rate as visit providers within that specialty. For example, 33.40% of medications ordered by gastroenterologists were discontinued by gastroenterologists and 26.93% were discontinued by cardiologists. Our interviews confirmed this finding; physicians working with “very specialty-specific medications that are critical and have very nuanced sigs” felt strongly that this should never be the case, and that they do not “want any other provider ever touching [their] medication list” without consulting the ordering provider. For other providers, it was important to review and update the medication list at each patient encounter if possible, and to discontinue all medications that were no longer being taken.

Discussion

We found that over 50% of medications requested for removal remained on the medication list, and only 28.56% of encounters without any removal requests involved discontinuation of medications in the EHR. This finding indicates that requests for medication changes that begin in the patient portal or during intake may not be a major factor influencing the decisions that physicians make regarding the discontinuation of medications. Rather, it appears that the specialty of the visit provider plays a greater role: depending on specialty, medications ordered by a provider from a particular specialty tend to be removed by a provider of the same specialty, possibly due to the fact that providers within a specialty may be able to manage critical patient medications more safely. Also, providers in specialties involving higher risk in medication list changes seem to be less likely to discontinue medications during an encounter. Further research into the accuracy of medication lists is necessary.

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